ROBERTO BLOOM

Geospatial Software Engineer · 3D Reconstruction & Remote Sensing · FAA Part 107
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SUMMARY

Geospatial software engineer building production drone-to-3D reconstruction pipelines — COLMAP Structure-from-Motion through Gaussian Splatting to SPZ-compressed 3D Tiles rendered in CesiumJS at true world coordinates. Delivered 0.24 m 3D georeferencing RMSE and 0.107 m horizontal GPS residual across 119/120 solved cameras. FAA Part 107 pilot who plans the mission and writes the software that turns it into a georeferenced product — a combination most candidates cover only half of.

TECHNICAL SKILLS

3D Reconstruction Gaussian Splatting · COLMAP (Structure-from-Motion) · Nerfstudio · photogrammetry · LiDAR point clouds · SPZ · 3D Tiles · RealityCapture · Jawset Postshot · CloudCompare · DJI Terra

Geospatial & Geodesy CesiumJS · QGIS · ArcGIS Pro · WGS84 / ECEF / ENU · Sim3 alignment · EXIF GPS alignment · geoid & terrain correction · coordinate reference systems · EPSG / CRS handling · PDAL · COPC · GeoJSON · RMSE accuracy QA

Languages & Platforms Python · TypeScript · React Native / Expo · Next.js · Flask · FastAPI · PHP · Three.js · Docker · AWS S3 · Git

UAS & Sensors FAA Part 107 · DJI Matrice / M3M · RTK mission planning · nadir & oblique capture · GSD targeting · KML/KMZ · multispectral (Sony IMX454) · thermal · RGB

EXPERIENCE

GIS Developer — **GeoInformatica Consulting** *(geospatial field-data software · contract, remote)* Jun 2026 – Present

- Architected an end-to-end pipeline converting raw drone imagery into georeferenced 3D models: COLMAP SfM, EXIF GPS-to-local-ENU Sim3 alignment, Nerfstudio Gaussian Splatting, SPZ-compressed 3D Tiles, and CesiumJS visualization at true latitude, longitude, and elevation.
- **Delivered 0.24 m 3D georeferencing RMSE and 0.107 m horizontal GPS residual** (119/120 solved cameras), exposed in-app as a per-reconstruction accuracy/QA readout.
- Resolved geospatial placement failures rooted in WGS84 ellipsoidal height versus MSL/geoid datum offsets and terrain-height correction, bringing models into correct real-world alignment.
- Built a human-in-the-loop supervised classifier for Gaussian scenes, applied across a 1.3M-Gaussian capture — operator QA taught it domain classes (including a dead-vegetation class no unsupervised cluster produced), exported to 3D Tiles and GeoJSON for ArcGIS.
- Shipped a real-world-anchored 3D viewer in the app's Reality Capture tab so field users open GPS-located models directly from drone imagery.
- Built drone-imagery ingestion — EXIF GPS extraction, map pinning, mission grouping, and offline-first storage — across React Native/Expo, TypeScript, Python, and PHP.

Reality Capture Engineer / Lead Drone Operator — **Firescore AI** *(wildfire analytics · remote + CA field ops)* Feb 2025 – May 2026

- Planned and flew UAV missions for wildfire inspection, vegetation analysis, parcel review, and 3D reconstruction across RGB, multispectral, thermal, and LiDAR-supported workflows.
- Authored repeatable capture standards for nadir and oblique missions — obstacle clearance, height tiers, image overlap, GSD targets, RTK, lighting consistency — making ad-hoc flights reproducible.
- Prototyped geospatial Gaussian Splatting workflows converting GPS-tagged imagery into CesiumJS-viewable 3D assets using COLMAP, EXIF-GPS-to-ENU alignment, parcel overlays, nearest-source-image lookup, and RMSE georeferencing QA.
- Developed and validated a Windows SDK application workflow for the Sony IMX454 multispectral sensor, covering sensor integration, capture modes, and vegetation-analysis review.
- Processed large drone datasets across DJI Terra, RealityCapture, CloudCompare, QGIS, ArcGIS Pro, and AWS S3, working with fire scientists, GIS specialists, and engineers to turn field constraints into repeatable workflows.

EARLIER EXPERIENCE

Inorganic Lab Technician — ALS Global. Environmental sample prep for TCLP analysis under SOP and documentation controls; eLIMS/GEL data capture. May – Oct 2024

Undergraduate Research Assistant — Trevor Nichols Research Center. Insecticide trials on fruit-crop pests; microscopy-based quantification for spotted-wing drosophila research. Summers 2022, 2023

EDUCATION & CERTIFICATION

B.S. Entomology — Michigan State University, East Lansing, MI Apr 2024

FAA Part 107 Remote Pilot Certificate · Valid Driver's License